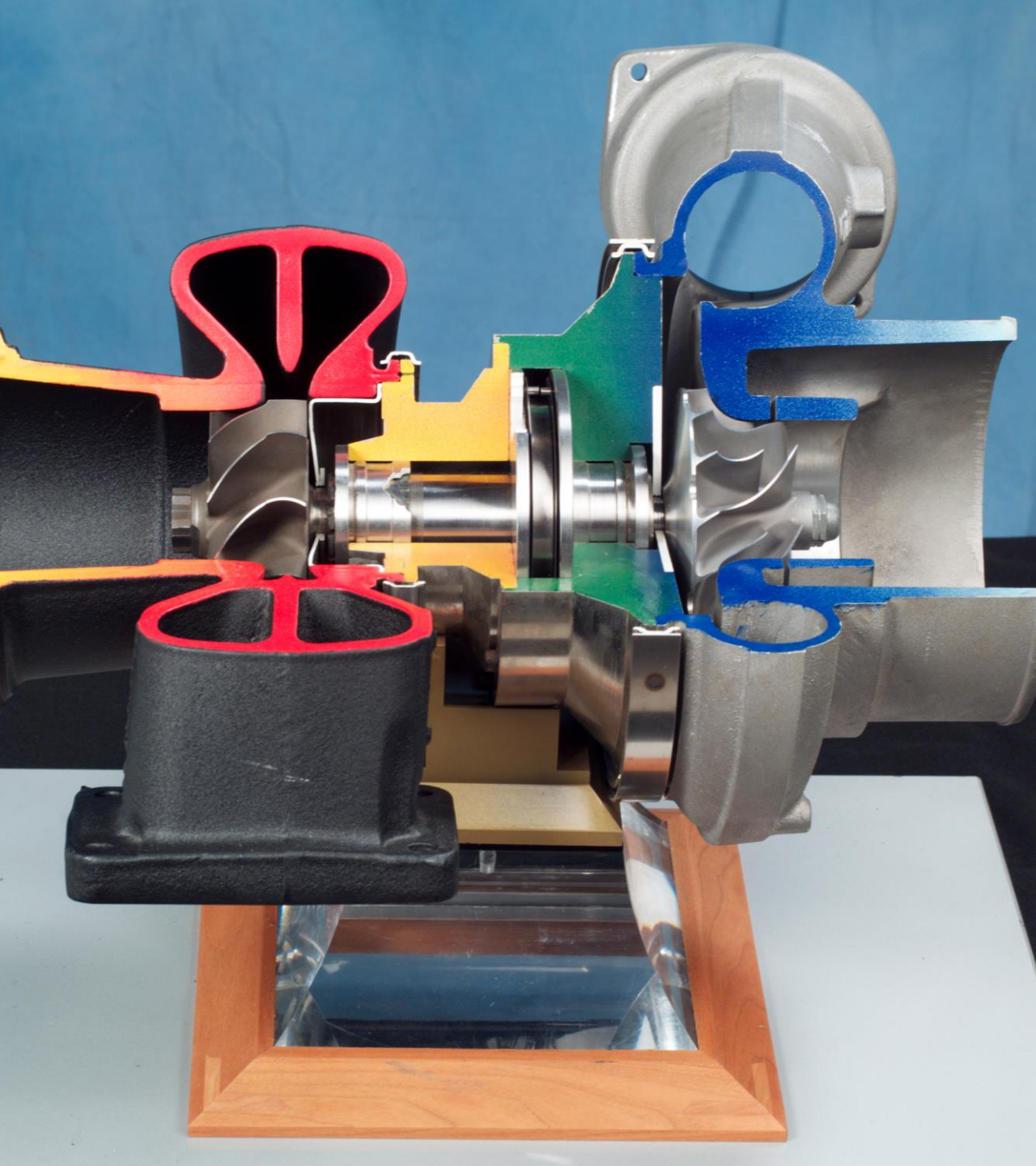




ME 300: Engineering Thermodynamics

Example: Isentropic Efficiency

Compressor example

Air enters a multistage compressor of a jet engine at 0.65 atm and 275 K with a flow rate of 63.5 kg/s. The pressure ratio of the compressor is 24:1 and it has an isentropic efficiency of 94%. Determine the power required to drive the compressor and the temperature of the air at the exit.

Given : $P_1 = 0.65 \text{ atm}$
 $T_1 = 275 \text{ K}$
 $\dot{m} = 63.5 \text{ kg/s}$
 $P_2/P_1 = 24.1$
 $\eta_{\text{isen}} = 0.94$

Find : $\dot{W}_{\text{comp}}$ ✓
 T_2 ✓

Assume : steady-flow
: $\Delta ke = \Delta pe = 0$
: $\dot{Q} = 0$
: ideal gas, const c_p

$$\Rightarrow \dot{m} \Delta h = -\dot{W}$$
$$\dot{W}_{\text{comp}} = -\dot{m} (h_2 - h_1)$$
$$= -\dot{m} c_p (T_2 - T_1)$$

isentropic

$$P_2 = 24 P_1$$
$$\dot{W}_{\text{ideal}} = -\dot{m} c_p (T_2 - T_1)$$
$$\eta_{\text{isen}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}}$$



What assumptions are appropriate for this problem?

- A. Steady flow
- B. $\Delta ke = \Delta pe = 0$
- C. Adiabatic
- D. All of the above

Energy conservation

Isentropic case

State 1: $T_1 = 275 \text{ K}$
 $P_1 = 0.65 \text{ atm}$

NOTE: The equation should be $T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$

State 2s: after isentropic compression by ideal compressor

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma}{\gamma-1}} \Rightarrow T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma}{\gamma-1}} \quad \gamma = 1.4$$
$$= 275 (24)^{1.4/0.4} = \underline{\underline{T_2 = 681.9 \text{ K}}}$$

ideal compression: $\dot{W}_{\text{ideal}} = -\dot{m}_G (T_{2,s} - T_1)$

$$= - (63.5)(1001)(681.9 - 275)$$
$$= - 25.8 \text{ MW}$$

Actual case

$$\eta_{isen,c} = \frac{\dot{W}_{ideal}}{\dot{W}_{real}} \Rightarrow \dot{W}_{real} = \frac{\dot{W}_{ideal}}{\eta_{isen,c}} = \frac{-25.8 \text{ MW}}{0.94}$$

$$\boxed{\dot{W}_{real} = -27.5 \text{ MW}}$$

$$\dot{W}_{real} = -\dot{m}c_p(T_2 - T_1)$$

$$T_2 = T_1 - \frac{\dot{W}_{real}}{\dot{m}c_p} = 275 - \frac{(-27.5 \times 10^6)}{(63.5)(1001)} \Rightarrow \boxed{T_2 = 707.7 \text{ K}}$$

$T_2 > T_{2s} \checkmark$